Stat 165 Final Project Write-Up

Background:

The goal of this paper is to study the effects of policy changes within the United States' largest domestic airline. Southwest has offered two free checked bags to customers since 1971, but starting May 28, 2025, this longstanding policy will no longer be implemented. Only A-List Preferred Members will receive their first and second checked bags for free.[1] Another change beginning on May 28th, are expiring flight credits. Starting in 2020, a Rapid Rewards member was able to accrue and save their flight credits, but now these credits will have "specified expiration dates".[2] These upcoming policy changes have received significant backlash from frequent leisure flyers and budget-conscious families. An article by Fox Business was flooded with comments from flyers who expressed their disappointment in the company, one commenter even stated, "Southwest is officially dead."[3] With growing backlash, competitor airlines such as American and Delta are expecting to see an increase in fliers who looked to Southwest for cheaper flight options. Free luggage meant Southwest offered cheaper flight fares for large families, but with the addition of baggage fares, Southwest is no longer the cheaper option, and families may begin flying with other large domestic airlines, like Delta or American. While we may see a decline in family travelers, Southwest will likely not see any changes in their weekend or one-night fliers. Overnight travelers often only have a carry on, and with baggage fares not directly affecting flight fares, these flights are still some of the best in the market.[3]

In 2023, Southwest revenue from checked bags was approximately 70 million dollars, which is significantly lower than other U.S. Airlines, such as American and United, who racked in over 1.2 billion dollars.[4] Baggage fees are expected to contribute 1.5 billion dollars to Southwest's annual revenue.[5] With changes to tariffs, and relatively low earnings since the start of the year, Southwest is hoping these changes will help restructure the company, but analyses have shown that ending their free-bags policy will lead to a "$1.8 billion in lost market share." [4] Southwest is known for their loyal customer base, who frequent the airline due to the no charge on checked luggage as well as other perks. Ryan Green, Southwest's former Executive Vice President and Chief Transportation Officer, believes "Changing our bags policy would be value destruction."[3] Given concerns from the public as well as executives within the company, we are curious how these new policies will affect Southwest. Will public backlash be strong enough to disrupt customer behavior on a scale that forces the company to reconsider or even reverse its decision? This paper will look through historical flight data in an attempt to uncover how

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[1] "Rapid Rewards A-List Preferred Members will receive their first and second checked bags for free." – Southwest, "Southwest Policy Changes," https://support.southwest.com/helpcenter/s/article/policy-changes
[2] Southwest, 'Southwest Policy Changes"
[3] Daniella Genovese, "Southwest Airlines risks losing customers over new bag policy, experts say", FOXBusiness, March 14, 2025, https://www.foxbusiness.com/lifestyle/southwest-airlines-risks-losing-customers-over-new-bag-policy-expert-says
[4] Alison Sider, Dawn Gilbertson, "Bags Will No Longer Fly free on Southwest Airlines", The Wall Street Journal, March 11, 2025, https://www.wsj.com/business/airlines/southwest-airlines-bag-policy-charge-basic-economy-9549218e
[5] Rajesh Kumar Singh, "Southwest Airlines shifts to paid baggage policy to lift earnings", Reuters, March 11, 2025, https://www.reuters.com/business/aerospace-defense/southwest-airlines-shifts-paid-baggage-policy-lift-earnings-2025-03-11

large-scale policy changes within the United States' largest domestic airline will affect their passenger volume.

All data used in this forecast was downloaded from The Bureau of Transportation Statistics - all data is publicly available.[6] The cleaned datasets used within the models, as well as our visualizations, can be found within our GitHub repository[7], which will be referenced throughout the paper.

Forecasting Question and Criteria:

This leads us to our forecasting question: Compared to 2024, will current changes to long standing Southwest Airline policies, beginning May 28th, 2025, decrease the number of total passengers in June 2025 as well as the end of the third quarter? Our probability ranges from 0 – 1, with a decrease in passengers corresponding to 1. The accuracy of our prediction will be calculated using the Brier/quadratic score: $S(p,\ event) = 1 - (p - 1_{event})^2$, where 1_{event} is an indicator based on whether passengers have decreased or not. If Southwest Airlines stops operations, shuts down, before June 28th, 2025 (one month after policy changes), then the probability that the total number of passengers decreases will be considered 1. The same criteria will be applied to the end of the third quarter this year.

Framework

To determine whether Southwest will see a decline in total passengers after implementing these new policies, we used the following framework. First, we found a base estimate using Fermi estimation. Next, we used reference class forecasting to compare Southwest's passenger data to similar airlines to find how introducing fees impacts passenger volume. We adjusted the reference class estimate by running a simulation to model the influence of public backlash. We fit a distribution to our data using the MLE and ran simulations based on this distribution to create a 95% confidence interval containing the mean number of passengers in June 2025 as well the third quarter: July, August, and September if there were no policy changes. Lastly, we combined our estimates to form our final forecast. We excluded data from the years 2002, 2020, 2021, 2022 and 2025. 2002 and 2025 were excluded because there's limited/incomplete data. 2020 and 2021 were excluded because they were impacted by COVID-19. 2022 was excluded because it's a transition year and is likely affected by COVID-19.

Key considerations

There are uncontrollable factors beyond these policy changes that could impact the total number of passengers flying Southwest, such as economic factors like inflation, the rising costs of needs, increased unemployment, and Trump's tariffs. The tariffs would increase travel costs, leading to higher business expenses because airlines would need to pay more for building materials, technology, catering, etc. To cover the increase in operating costs, they likely would

[6] All data was accessed from The Bureau of Transportation Statistics, https://www.transtats.bts.gov/Data_Elements.aspx?Qn6n=E

[7] "Stat-165-Forecasting-Project", https://github.com/SuperNova707/Stat-165-Forecasting-Project/tree/main

increase their ticket prices. With consumers having less disposable income and costs rising, the total number of passengers would decrease.

Political factors with limited impact are stricter immigration policies and increased deportations, which decrease U.S. travel by inciting fear. This would impact the total number of passengers who fly Southwest's limited international destinations. It can also decrease domestic travel by international students and other visa holders who fear facing more aggressive TSA officers who could report them to CBP. Recent examples of these events are the UCLA student who was detained at the U.S.-Mexico border[8] and the termination of student visas across U.S. universities, including international students from UC Berkeley.[9]

To account for outside influences, we used Frontier, Spirit, JetBlue, and Alaska as reference classes. We selected these airlines because they are domestic U.S. airlines with limited international options, are Southwest's competitors and serve similar customer demographics, and have introduced baggage fees in the past. JetBlue and Alaska began charging for checked bags in June 2015[10] and May 2010[11] respectively, while Spirit and Frontier began charging for carry-on bags in August 2010[12] and April 2014[13] respectively. Comparing the trends for each airline before and after policy implementation, we can infer if the change in passengers is due to outside factors or the policy change. Based on how policy changes impacted our reference classes, we can form a base estimate on the probability that Southwest sees a decline in total passengers in June 2025 and at the end of the third quarter.

Another unpredictable factor is public backlash since Southwest's brand identity is tied to providing free checked bags. To account for this, we simulated various levels of public backlash and the impact each level could have on passenger volume, then updated our base estimate.

Forecasting Skills and Simulations

All figures referenced throughout this section can be found within the GitHub following the file path: */main/plot-visualizations/*. Each figure is labeled by order in which they are referenced (e.g., Figure A Jetblue.png will be referred to as Figure A).

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[8] Austin Turner, "UCLA Student Detained at U.S.-Mexico Border," *KTLA*, April 18, 2025, https://ktla.com/news/local-news/international-ucla-student-detained-at-u-s-mexico-border/

[9] Doha Madani and Lindsay Good, "Visas Revoked for More Than 3 Dozen California University Students and Alumni," *NBC News*, April 6, 2025, updated April 7, 2025, https://www.nbcnews.com/politics/immigration/visas-revoked-california-university-students-alumni-rcna199905

[10] Ben Mutzabaugh, "JetBlue Details New Bag Fees, Which Begin Tuesday," *USA Today*, June 30, 2015, https://www.usatoday.com/story/todayinthesky/2015/06/30/jetblue-details-new-bag-fees-which-begin-tuesday/29503705/

[11] Alaska Airlines, "Alaska Airlines and Horizon Air Announce Service Fee Changes" *Alaska Airlines Newsroom*, April 22, 2010, https://news.alaskaair.com/newsroom/alaska-airlines-and-horizon-air-announce-service-fee-changes/

[12] ABC News, "Spirit Airlines to Charge New Fee for Carry-On Luggage" *ABC News*, April 6, 2010, https://abcnews.go.com/Travel/spirit-airlines-charge-fee-carryon-luggage-bag-surcharge/story?id=10298802

[13] ABC News, "Frontier Airlines Will Charge Some Customers for Carry-Ons, Introduces a Fee for Soft Drinks" *ABC News*, May 2, 2013, https://abcnews.go.com/Travel/frontier-charge-carry-bags-soda/story?id=19091788

Fermi Estimates - Prediction & Reasoning:

The net income of each airline has been decreasing post-COVID, as shown in Figure 1. Southwest has been in a steady linear decline since 2016 (ex-COVID) and is approaching an annual net income of break even.

As depicted in Figure 2, Southwest with their acquisitions of AirTran in 2011 have increased their passengers and flights significantly. Alaska, Frontier, JetBlue, and Spirit have seen only modest growth in passengers and flights in the last 20 years. Despite the falling net income, operating revenue for the airlines has been increasing due to the increasing passenger numbers. All airlines, except JetBlue and Southwest, first started introducing checked bag fees in 2008, likely due to the Great Recession, in an attempt to recover revenue lost due to falling passenger numbers at that time. The overall trend in the airline industry over the last year, has been declining operating revenue, illustrated in Figure 3, which is causing airlines to look for additional sources of income.

For Southwest, starting in the early 2000s, the number of passengers per flight was 75, and has steadily increased to the current average of approximately 120-125 passengers per flight. However, other airlines are able to get 150-160 passengers per flight hence generating more revenue per flight. Frontier is able to get 175 passengers per flight, yet still charges for checked bags and carry-ons, as well as seats. The other airlines also charge for checked bags and seats, which is becoming a revenue generating norm among US airlines - see Figure 4 for side by side comparison of Annual Passengers per Flight.

The revenue per passenger for Alaska and JetBlue is significantly higher than Southwest and the other budget airlines, as seen in Figure 5. Alaska and JetBlue have additional revenue from their higher class fares and baggage fees, whereas Southwest has not captured that revenue yet and will be introducing those charges at the end of May to increase revenue.

Fermi Estimates - Comparison to JetBlue, Other External Factors in the Aviation Industry

JetBlue introduced baggage fees in 2015, and did not see any drop in operating revenue or in the number of passengers.

By 2015, baggage fees have already been the norm in the US airline industry for 7 years, thus passengers have already been accustomed to paying those fees on every flight. Even though JetBlue had financial hardships from 2004-2010 due to rising fuel prices and other factors, their customer base remained loyal and rated the airline high in customer satisfaction.

The aviation industry was significantly rocked beginning in 2024 by quality control issues at Boeing returned to light after a door plug blew off an Alaska Airlines Boeing 737 MAX 9 jet, as a result, the aircraft giant had to scale back production in order to make sure the planes they built were tested and safe. Because Southwest Airlines exclusively flies the Boeing 737, and they were unable to fly as many passengers as they can due to the lower capacity and delivery delays from Boeing, the airline is already losing passengers as a result[14], which is about a 1.67% (0.0167) decline.

Reference Class Forecasting and Public Backlash Simulation:

To account for the impact of uncontrollable economic and political factors on our forecast, we filtered the dataset containing the total passengers by airline to contain only the data for May and June and grouped by airline, year, and month. For estimating the outcome for May 2025 to June 2025, we made four individual data frames for each reference class that contained the passenger data after each respective baggage policy was implemented. To compare the change in passenger volume after policy implementation, we found the percent change from May to June for each year. We plotted bar graphs (Figures A, B, C, and D) to observe the trend for each airline. We found that since making the policy change, over the years JetBlue experienced a decline in number of passengers 2 out of 7 times. Spirit experienced this 3 out of 11 times, Frontier experienced this 0 out of 8 times, and Alaska experienced this 0 out of 12 times. From 38 total observations, we saw a decline in the total number of passengers 5 times. We used this to form the base estimate that the chance of a decline in total passengers flying Southwest from May 2025 to June 2025 is $\frac{5}{38} = 0.1316$.

To account for the impact of public backlash on our forecast, we simulated various levels of backlash (low, medium, high), assigned each level our belief of the probability it occurs, and how much we expect that level of backlash to impact our base estimate that we found through reference class forecasting. We said there's a 60% chance of little backlash, 30% chance of moderate backlash, and 10% chance of major backlash. We think low backlash will have no effect on the base estimate, medium backlash will increase it by 0.05 and high backlash will increase it by 0.10. We sampled from the three scenarios 10,000 times and applied each effect to our base estimate, resulting in 10,000 sample estimates. To find our adjusted estimate, we took the average of our sampling distribution (Figure E). We found our updated estimate to be 0.1566.

We followed similar steps to find an estimate for the total passengers at the end of the third quarter in 2025. Instead of finding percent change, we found the total number of passengers in the third quarter for each airline in each year. We plotted this data (Figures F, G, H, I) and observed an upward trend for all reference classes. Following each respective policy change, the total passengers in the third quarter increases every year. This suggests that implementing baggage fees doesn't lead to a decline in total passengers at the end of the third quarter. We found our base estimate using Laplace's rule of succession, which states that if an event has *n* opportunities to occur but has never happened, assign the probability $\frac{1}{n+2}$ to it happening the next time. From our 4 instances of policy changes among different airlines, we didn't see a

[14] Osipov, D. S., "Southwest Airlines lost nearly 25,000 passengers at this airport", Simple Flying, April 4, 2025, https://simpleflying.com/southwest-airlines-lost-25000-passengers-this-airport/

decline in total passengers at the end of the third quarter. Therefore, our base estimate that Southwest experiences a decline in total passengers at the end of the third quarter after changing policies is $\frac{1}{4+2} = \frac{1}{6} = 0.1667$. To account for public backlash, we followed the exact same steps and found our updated estimate to be 0.1917. Our sampling distribution can be seen in Figure J.

Fitting the data using AIC/BIC scores and creating a 95% confidence interval with MLE:

The goal for this experiment was to attempt to identify what distribution Southwest's total flier volume follows in June as well as in their third quarter of each year. As discussed earlier in the paper, certain years were excluded in order to ensure the data was consistent. The coding was done in R using MASS and fitdistrplus packages. We initially plotted the true data to see if there were any outliers or unusual patterns within the data. Then the data was fitted against three likely distributions: Normal, Lognormal, and Poisson, and each of their AIC and BIC scores were calculated. The log normal distribution had the lowest for both scores, so we fitted the data using this distribution. We also shifted the data using quadratic weights, following the assumption that early 2000's data should affect the outcome less than say passenger volume in 2024. We initially tested linear and exponential weights, but concluded that they did not correctly express the importance of the data.

Using the weighted log normal distribution, we linearly regressed the model and predicted the best fit mean and standard error. We then calculated the MLE using the expectation of a log normal distribution and using 10,000 simulations we created a 95% confidence interval for the likely mean passenger volumes, assuming no policy changes. For June 2025, our confidence interval was [15,737,644, 18,717,418], for the third trimester it was [44,750,886, 53,009,371] which based off of 2024's data of 16,402,763 and 44,711,432 respectively, seem to be a reasonable forecast - all code for this section can be found within our main qmd file.[7]

Of course, we understand the limitations of this method considering we are only working with 20 data points. AIC and BIC are known to be less reliable with small sample sizes, but despite this, our log-normal model appears to be a relatively good fit, and the constructed 95% confidence interval for the mean expected passage volume is consistent with our initial data.

Our Prediction for Southwest

Southwest has a lower passenger per flight number compared to the other airlines, and a decreasing net income. The airline needs to do something to boost profitability. Southwest's ticket prices are not the cheapest in the airline industry, and by charging for bags, the overall price to fly with Southwest is going to be even more expensive.

Based on the facts above, we predict that Southwest will see around a 7.4% decline in passengers in June 2025, and around 10.4% decline by the end of Q3 2025 (including June), as a result of baggage fees the airline will introduce at the end of May. In regards to June 2025, we have concluded after analyzing a variety of factors, that there is a 60% chance Southwest will have fewer fliers than in June 2024. For the third quarter, we believe that there is a 40% chance Southwest will see a decline in passenger volume in comparison to 2024. While changes to policies may cause shockwaves, charging fees are the norm for all other airlines, so we anticipate over the next 12 months, Southwest will slowly recover the passengers lost due to the new fees as customer loyalty returns to the airline, and passengers get used to the new charges.